

Complete Beginner Guide: Ubuntu Host → Windows VM with KVM

This guide combines:

- the **first KVM guide**
- the **Output-2.pdf VM creation steps**
- the **missing practical steps**
- common errors and fixes

Goal:

Ubuntu Linux



Install KVM



Create Windows VM



Install Windows



Run a Hello World program

PHASE 1 — Prepare Ubuntu

Step 1 — Verify virtualization support

Open Terminal and run:

```
cat /proc/cpuinfo | grep -Ec '(vmx|svm)'
```

Expected:

1

or higher.

Meaning:

- vmx = Intel VT-x
- svm = AMD-V

Possible Errors

Output is:

0

Cause

Virtualization disabled in BIOS.

Fix

Reboot → enter BIOS/UEFI → enable:

- Intel VT-x
OR
- AMD-V / SVM

Step 2 — Update Ubuntu

```
sudo apt update && sudo apt upgrade -y
```

Possible Errors

“Could not get lock”

Another package manager is running.

Fix

Wait or run:

```
sudo killall apt apt-get
```

carefully.

Step 3 — Install KVM packages

Run:

```
sudo apt install -y \
```

```
qemu-kvm \
```

```
virt-manager \
```

```
libvirt-daemon-system \
```

```
virtinst \
```

```
libvirt-clients \
```

```
virt-viewer \
```

```
ovmf \
```

```
swtpm \
```

qemu-utils

What these do

Package	Purpose
qemu-kvm	virtualization engine
virt-manager	GUI VM manager
libvirt	VM service
ovmf	UEFI support
swtpm	TPM support
virt-viewer	better VM display

Possible Errors

“Unable to locate package”

Usually:

- internet issue
- outdated package lists

Fix

Run:

```
sudo apt update
```

then retry.

Step 4 — Enable libvirt

Run:

```
sudo systemctl enable --now libvirtd
```

Verify:

```
sudo systemctl status libvirtd
```

Expected:

active (running)

Press:

q

to exit.

Possible Errors

“Unit libvirtd.service not found”

Very uncommon on Ubuntu.

Fix

Reinstall libvirt packages.

Step 5 — Add user permissions

Run:

```
sudo usermod -aG kvm $USER
```

```
sudo usermod -aG libvirt $USER
```

Step 6 — Reboot

IMPORTANT.

```
sudo reboot
```

Without reboot/logout:

- Virt-Manager may fail permissions.
-

Step 7 — Verify KVM works

After reboot:

```
virsh list --all
```

Expected:

```
Id Name State
```

```
-----
```

No permission errors.

Possible Errors

“permission denied”

Fix

You forgot reboot/logout.

PHASE 2 — Download Windows

Step 8 — Download Windows ISO

Download from Microsoft.

Recommended:

- Windows 10 ISO

Save somewhere easy:

~/Downloads/Win10.iso

Optional but recommended

Download:

- VirtIO driver ISO

Search:

virtio-win.iso

This improves:

- disk speed
 - networking
 - VM responsiveness
-

PHASE 3 — Create the VM (from PDF)

Step 9 — Open Virt-Manager

Open app launcher:

Virtual Machine Manager

Expected:

QEMU/KVM — Connected

Possible Errors

“Not connected”

Fix

Run:

```
sudo systemctl restart libvirtd
```

Step 10 — Create new VM

Click:

Create a new virtual machine

(from Output-2.pdf)

Step 11 — Select install media

Choose:

Local install media (ISO image or CDROM)

Click Forward.

Step 12 — Select Windows ISO

Browse and select:

Win10.iso

Uncheck:

Automatically detect from installation media

if detection fails.

Set OS manually:

- Microsoft Windows
 - Windows 10
-

Possible Errors

ISO not booting

Cause:

- corrupted ISO

Fix:

- re-download ISO
-

Step 13 — Assign RAM and CPUs

Recommended:

Setting Value

RAM 4096–8192 MB

CPU 2–4

(from PDF)

Step 14 — Create virtual disk

Recommended:

OS	Disk Size
----	-----------

Windows 10	50 GB
------------	-------

Choose:

Create disk image

Possible Errors

“No space left”

Host SSD full.

Fix:

- free space
 - move VM storage
-

Step 15 — Before Finish (IMPORTANT)

Before clicking Finish:

CHECK:

Customize configuration before install

This missing step is important.

Step 16 — Enable proper firmware

Inside configuration:

Set:

Firmware → UEFI x86_64

(OVMF)

This improves Windows compatibility.

Step 17 — Add TPM (for Windows 11 only)

If using Windows 11:

Add Hardware →

- TPM
 - Emulator
-

Step 18 — Begin installation

Click:

Begin Installation

Now Windows installer starts.

PHASE 4 — Install Windows

Step 19 — Normal Windows setup

Proceed normally:

- language
 - keyboard
 - install now
-

Possible Error — “No drives found”

Very common.

Fix using VirtIO drivers

Add VirtIO ISO

In VM window:

View → Details

Add second CD-ROM:

- attach virtio-win.iso

During Windows setup

Click:

Load Driver

Browse:

viostor

Then disk appears.

Step 20 — Finish Windows installation

Windows reboots several times.

Eventually you reach Windows desktop.

PHASE 5 — Improve VM usability

Step 21 — Install VirtIO guest tools

Inside Windows:

open VirtIO ISO.

Install:

- network drivers
 - storage drivers
 - QXL/SPICE drivers
-

Step 22 — Install SPICE guest tools

This enables:

- clipboard sharing
 - auto resize
 - smoother mouse
-

PHASE 6 — Run a Hello World program

Option A — Python (easiest)

Inside Windows VM:

Download Python.

Create:

```
print("Hello World")
```

Run:

```
python hello.py
```

Expected:

Hello World

Option B — C Program

Install:

- Visual Studio
OR
- MinGW

Program:

```
#include <stdio.h>
```

```
int main() {  
    printf("Hello World");  
    return 0;  
}
```

Compile:

```
gcc hello.c -o hello
```

Run:

hello.exe

Common Problems Summary

Problem	Cause	Fix
virtualization disabled	BIOS setting	enable VT-x/SVM
Virt-Manager not connected	libvirtd stopped	restart service

Problem	Cause	Fix
permission denied	forgot reboot	reboot
no disks in Windows setup	missing VirtIO drivers	attach VirtIO ISO
VM extremely slow	low RAM/CPU	increase allocation
no internet in VM	libvirt network issue	virsh net-start default
mouse trapped	missing SPICE tools	install guest tools

Optional Networking Fix

If VM has no internet:

Run on Ubuntu host:

```
sudo virsh net-start default
```

```
sudo virsh net-autostart default
```

Final Result

You end up with:

Ubuntu host



Virt-Manager



Windows VM



Run programs normally

You can:

- fullscreen Windows
- minimize it
- pause it
- clone it
- delete it
- run multiple VMs later

That's the complete practical flow.

